

Tanuj Kumar

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Education

University of Wisconsin-Madison	Fall 2021 - *
Ph.D., Department of Electrical & Computer Engineering	4.0/4.0
Advisor: Dr. Mikhail Kats	
Birla Institute of Technology & Science (BITS), Pilani, India	2015 - 2020
Integrated Dual Degree program	7.96/10
Bachelor of Engineering (Honours) in Electrical and Electronics	
Master of Science (Honours) in Chemistry	

Awards

1. Student Research Grants Competition (SRGC) Research Travel Award, University of Wisconsin-Madison, 2022
2. ECE Chancellors Opportunity Fellowship (COF) award, University of Wisconsin-Madison, 2021

Journal Publications

1. **T. Kumar**, R.E. Owyung, S. Sonkusale, *Rapid Cleanroom-Free Fabrication of Thread Based Transistors Using Three-Dimensional Stencil-Based Patterning*, Flexible and Printed Electronics, Feb 2021. [\[DOI\]](#)

Patents

1. S. Sonkusale, R.E. Owyung, **T. Kumar**, *Fabrication of Thread Based Transistors*, Provisional patent application filed Jan 2021.

Conference Presentations

1. **T. Kumar**, D. Feng, S. Yin, P. Lin, M. Mah, M. Fortman, G. Jaffe, C. Wan, C. Fang, R. Warzoha, V. Brar, J. Talghader, M. Kats, *Photothermal Commonpath Interferometry of Silicon Nitride Membranes for Laser Light Sails*, Conference on Lasers and Electro-Optics (CLEO), May 2023 (accepted).
2. **T. Kumar**, A. Sugathan, K.L. Narasimhan, A. Pandey, S. Avasthi, *CuFeSe₂ Quantum-Dot Based Infrared Photodetectors with Functionality in the Ambient* (Poster), IEEE Sensors Conference (online), Nov 2021. [\[DOI\]](#)
3. **T. Kumar**, R. E. Owyung, S. Sonkusale, *High Throughput Cleanroom-Free Fabrication of Thread Based Transistors*, Materials Research Society Fall Meeting (online), Nov 2020.
4. S.M. Raje, A. Goel, S. Sharma, K. Aggarwal, D. Mantri, **T. Kumar**, *Development of On Board Computer for a Nanosatellite*, International Astronautical

Congress, Sep 2017. [\[PDF\]](#)

Talks

1. *Building a Satellite Isn't Rocket Science*, TEDx BITS Pilani, India, Jan 2019 [\[Link\]](#).
2. *Role of Open Source in the Development of a Student-Built Nanosat & its Team Members*, Open Source Cubesat Workshop - European Space Agency, Madrid, Sep 2018 [\[Link\]](#).

Research Experience

1. Characterization of low-loss laser-sail materials Sep 2021 - ongoing
University of Wisconsin-Madison - with Prof. Mikhail Kats
We are designing and conducting experiments to find the optical parameters of very low-loss materials. These materials have applications in laser sails, to be used in the Breakthrough Starshot project.
2. Quantum Dot Photodetectors Aug 2020 - Jul 2021
Indian Institute of Science - with Prof. Sushobhan Avasthi
We designed the fabrication flow and materials for ambient-operation broadband CuFeSe₂ and CuFeS₂ quantum dot photodetectors.
3. Rapid Fabrication Process for Thread-Based Transistors Aug 2019 - June 2020
Tufts University - with Prof. Sameer Sonkusale
We developed a high-throughput, cleanroom-free fabrication method for Thread-Based Transistors (TBTs). This improved on the throughput of a previous fabrication method by 2 orders of magnitude.

Internships

1. CentraleSupélec, Paris May 2019 - July 2019
With Prof. Caroline Lelandais-Perrault
2. Indian Space Research Organization May 2018 - July 2018
With Dr. Vinod Kumar, FIE, FIETE, Scientific Officer
3. Indira Gandhi Centre for Atomic Research May 2017 - July 2017
With Mr. Sanga Ramesh, Scientific Officer

Teaching Assistantships (During Undergraduate)

1. Microelectronic Circuits (EEE F244) Jan 2019 - May 2019
 - Designed and conducted tutorial sessions on SPICE & Cadence simulation techniques.
 - Evaluated students' performances in tests.
 - Group coordinator for my TA group.
2. Computer Programming (CS F111), Aug 2018 - Dec 2018
 - Conducted twice-a-week lab classes on Linux commands and C programming

for freshman year students.

- Developed problem sets and answer keys.
- Invigilated course quizzes.